

RESEARCH ARTICLE

Sustainable Product Innovation Through Horizontal Coopetition: The Role of Boundary-Spanning Search and Partner Similarity

Xia Wu¹  | Jianjun Yang¹ | Sandra Hasaneftendic² | Bart Bossink²

¹School of Management, Xi'an Jiaotong University, Xi'an, Shaanxi, China | ²Breakthrough Tech Innovation Group, Faculty of Science, Vrije Universiteit Amsterdam, Amsterdam, the Netherlands

Correspondence: Jianjun Yang (jjyang@mail.xjtu.edu.cn)

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ABSTRACT

This study is the first to examine the mediating role of boundary-spanning search and the moderating role of partner similarity in horizontal coopetition for sustainable product innovation. Analyzing empirical data from 236 Chinese firms, we found that horizontal coopetition positively impacts both proactive search (aimed at surpassing competitors) and responsive search (aimed at following competitors), with a stronger influence on proactive search. Both search types mediate the relationship between horizontal coopetition and sustainable product innovation. Furthermore, greater competing similarity (e.g., market coverage) between firms and their partners increases firms' inclination toward proactive search under the influence of horizontal coopetition but weakens their tendency for responsive search. Conversely, greater cooperating similarity (e.g., corporate culture) increases firms' inclination toward responsive search under the influence of horizontal coopetition but weakens their tendency for proactive search.

1 | Introduction

The global surge in industrialization, population growth, and increased production has spurred economic development but has also caused significant environmental degradation (Severo et al. 2017; Hussain et al. 2020; C. H. Wang and Juo 2021). As the main creators of carbon emissions, commercial enterprises face the challenge of mediating environmental concerns while pursuing economic development, most commonly through sustainable product innovation (SPI) (Severo et al. 2017; Meiros et al. 2022; Calik 2024). SPI refers to balancing economic, environmental, and social goals in the process of product creation and improvement, minimizing negative environmental impact

while gaining economic benefits (Severo et al. 2017; Meiros et al. 2022; Calik 2024). However, its complexity and uncertainty often need collaborative relationships with various stakeholders (Mousavi et al. 2019; Meiros et al. 2022), including cooperation with direct competitors, which is also known as horizontal coopetition (Le Roy et al. 2022; Xie et al. 2023).

According to coopetition theory, coopetition refers to the co-existence of competition and cooperation between enterprises. Horizontal coopetition refers specifically to the cooperation between two enterprises at the same stage of the value chain to jointly create, produce, and sell new products while competing with other products in the same market (Le Roy et al. 2022;

Huo et al. 2024). Unlike other types of collaboration, horizontal coopetition means that companies share similarities with their partners in technology or product positioning, which makes it easier to increase opportunities for mutual competence growth in the process of knowledge exchange (Massari and Giannoccaro 2021; Le Roy et al. 2022; Xie et al. 2023). This collaboration can also share the risks and costs of research and development (Massari and Giannoccaro 2021). For example, although Toyota and BMW are direct competitors in the automotive market, they have established partnerships to jointly develop hydrogen fuel cell technology for drive systems. However, the inherent coopetition paradox implies that competition may lead to opportunism, knowledge leakage, and power imbalances (Bouncken and Kraus 2013; Le Roy et al. 2022; Corbo et al. 2023; Huo et al. 2024). Therefore, the answer to the question of how the value of horizontal coopetition for SPI can be controlled, increased, and improved becomes a critical issue.

Although scholars argue that coopetition is linked to firms' sustainability (Planko et al. 2019; Munten et al. 2021; Corbo et al. 2023; Mwesummo et al. 2023), additional mechanisms are still required in horizontal coopetition to improve SPI (Mwesummo et al. 2023). Specifically, the tension in horizontal coopetition creates explicit competitive pressure even during collaboration (Raza-Ullah and Kostis 2020; Czakon and Czernek-Marszałek 2021; Huo et al. 2024). Faced with competitive pressure, companies must continuously span organizational boundaries to search for diverse and valuable resources, providing new insights and opportunities for innovation (Li et al. 2024). Boundary-spanning search involves sourcing heterogeneous resources from diverse external sources (J. Wang et al. 2020; Yang et al. 2021; Li et al. 2024). Specifically, proactive search is directed toward gaining a competitive edge by staying ahead of rivals, whereas responsive search is aimed at following them (J. Wang et al. 2020; Yang et al. 2021; Li et al. 2024). Both proactive search and responsive search serve as sources of heterogeneous innovation resources for firms (Yang et al. 2021; Lee et al. 2024; Li et al. 2024) and may lay a solid foundation for SPI. Relevant questions that arise are: Does horizontal coopetition impact proactive search and responsive search? If so, are there differences in impact? And does this impact on these two search modes also affect firms' SPI? In summary, this study will examine whether proactive and responsive search each serve as significant intermediary pathways in the relationship between horizontal coopetition and SPI.

Furthermore, partner similarity, including competing and cooperating similarity, influences the behavior of collaborating enterprises and the outcomes of the collaboration (Kim and Parkhe 2009; Luo and Deng 2009; Gao et al. 2017). Therefore, we substantiate that partner similarity may be a contingent factor affecting the functioning of horizontal coopetition. Specifically, competing similarity refers to similarity in geographic market coverage, product market coverage, and areas of technological strength among partners (Kim and Parkhe 2009), possibly triggering a 'learning race', where firms gain competitive advantages only through being better at accumulating competencies than their partners (Bouncken and Kraus 2013; Gao et al. 2017). Conversely, cooperating similarity refers to similarity in corporate culture and management practices (Kim and Parkhe 2009), which is beneficial for promoting coordination of

the collaboration (Kim and Parkhe 2009; Vaez-Alaei et al. 2022). Resource-advantage theory suggests that competition is a dynamic process. This means firms typically make different strategic choices based on the competitive environment to gain relative competitive advantages (Hunt and Morgan 1996). Thus, in the context of horizontal coopetition, a central question is: Do different degrees of cooperating similarity and competing similarity affect firms' preference for different types of boundary-spanning search? Overall, this paper will further explore whether and how partner similarity affects the relationship between horizontal coopetition and boundary-spanning search.

Given the above, the contribution of this study is threefold. First, this study refines the research on coopetition in the field of firms' sustainability by exploring the impact of horizontal coopetition on SPI. Second, this study shows a new pathway for companies to achieve SPI through horizontal coopetition from the perspective of boundary-spanning search. Third, this study reveals the differentiated moderating mechanisms of partner similarity in the relationship between horizontal coopetition and boundary-spanning search and further deepens the research on the process mechanisms through which companies achieve SPI through horizontal coopetition.

The remainder of this paper is structured as follows. Section 2 discusses the theoretical background of the research model constructed in this study and the formulation of the research hypotheses related to it. Section 3 introduces the empirical research methods used to test the hypotheses. Section 4 reports, and Section 5 analyzes the survey research outcomes and discusses the contributions, limitations, and directions for future research. Finally, Section 6 concludes with the main answers to the research questions and their implications.

2 | Theoretical Framework and Research Hypotheses

2.1 | Theoretical Framework

2.1.1 | Coopetition in Firms' Sustainability

Scholars have recently emphasized the important role of coopetition in achieving sustainability in enterprises (Vu and Fath 2024; Mwesummo et al. 2023; Xin et al. 2023). Vu and Fath's (2024) empirical study on the New Zealand alternative protein industry shows that enterprises can jointly produce sustainable solutions through coopetition, which can not only exceed the market growth created by individual enterprises but also generate extensive social benefits. The research results of Mwesummo et al. (2023) show that coopetition is conducive to cultivating the dynamic capabilities of enterprises (such as knowledge development, organizational responsiveness, reflexive control, and fostering sustainability awareness), thereby improving the social, environmental, and economic dimensions of enterprise sustainability performance. Mirzabeiki et al. (2023) argue that coopetition is conducive to reducing logistics costs and carbon footprint, enhancing regional economic competitiveness, and reducing emissions to affect sustainability performance. In addition, some empirical studies have confirmed the positive impact of coopetition on carbon footprint reduction

(Rosa et al. 2024) and sustainable competitive advantage (Della Corte and Aria 2016).

By reviewing the literature, we found that existing studies mostly focus on the impact of coopetition on the overall sustainable performance of enterprises from a macro perspective (Vu and Fath 2024; Mwesummo et al. 2023; Della Corte and Aria 2016), lacking research on the role of horizontal coopetition in more specific forms of innovation such as SPI. Given that SPI necessitates technological exchange and knowledge integration, and that horizontal coopetition promotes a diverse exchange of ideas, knowledge, and resources among entities within the same industry or market (Le Roy et al. 2022), this study specifically examines the impact of horizontal coopetition on SPI, reveals the mediating path of horizontal coopetition on SPI, and identifies the contingent factors that affect the role of horizontal coopetition.

2.1.2 | Boundary-Spanning Search

Boundary-spanning search involves sourcing heterogeneous resources and information from diverse external sources such as customers, suppliers, research institutions, and universities (J. Wang et al. 2020; Yang et al. 2021; Li et al. 2024). It generally shows positive effects on the likelihood of generating highly valuable innovations (Park et al. 2024). Regarding the division of boundary-spanning search, existing literature has delineated dimensions of it from different perspectives, such as the depth and scope of search (S. S. Xiao et al. 2023), search broadness and search narrowness (Li et al. 2019), and interactive and non-interactive search (Roper et al. 2017). However, as noted by Katila and Chen (2008), these classifications primarily focus on the company itself and overlook the situation in which firms' search activity usually consider competitors' strategies and strategic maneuvers. On the one hand, companies pursue boundary-spanning search to surpass competitors and gain a first-mover advantage, which is proactive search (Katila and Chen 2008; J. Wang et al. 2020; Yang et al. 2021). On the other hand, companies engage in boundary-spanning search to follow and imitate competitors in order to reduce risks, which is known as responsive search (Katila and Chen 2008; J. Wang et al. 2020; Yang et al. 2021). To gain a deeper understanding of the behavioral logic of boundary-spanning search in competitive environments, this study categorizes it into proactive search and responsive search based on their objectives, drawing on the works of Katila and Chen (2008) as well as Yang et al. (2021).

At present, scholars have conducted much research on the impact of boundary-spanning search on organizational innovation, such as business model innovation (Zhu et al. 2023), innovation performance (Li et al. 2024), and new product innovation (Lee et al. 2024). However, except for studies by Shi et al. (2024) and Zhu et al. (2023), the antecedents of boundary-spanning search remain underexamined. As Shi et al. (2024) pointed out through meta-analysis, there is a gap in relevant research on the antecedents of boundary-spanning search. In particular, few scholars have explored the role of horizontal coopetition in promoting boundary-spanning search. Therefore, the link between horizontal coopetition and boundary-spanning search merits further research.

2.1.3 | Coopetition Theory

The concept of coopetition was initially coined in the 1980s by Ray Noorda, the founder of Novell, and has since been enriched and developed by scholars in the field of strategic management (Xie et al. 2023). The essence of horizontal coopetition lies in the coexistence of these contradictory logics of cooperation and competition, with the overriding motivation for firms to engage in coopetition to be able to create value (Gnyawali and Ryan Charleton 2018). Coopetition theory suggests that value creation through coopetition involves different mechanisms. On the one hand, focal firms can unilaterally learn by acquiring skills and knowledge from their alliance partner and applying them to activities unrelated to the alliance activities and outside its boundary (Rai 2016; Rai and Surana 2022). This implies companies may obtain complementary resources and critical technologies needed for SPI from horizontal coopetition. On the other hand, focal firms create value by competing against their alliance partner to outperform it in activities directly linked to the alliance-related activities but taking place beyond the alliance boundary (Rai 2016; Rai and Surana 2022). This implies that firms may actively seek opportunities driven by the inherent competitive attributes of horizontal coopetition, crossing organizational boundaries to search for heterogeneous resources to support innovation. In summary, coopetition theory is not only the theoretical foundation for our study on the mechanism by which horizontal coopetition affects SPI but also supports the proposal of boundary-spanning search as the mediating path between them.

2.1.4 | Resource-Advantage Theory

Resource-advantage theory, proposed by Shelby Hunt and Robert Morgan in the 1990s, is a general competitive theory that describes the evolutionary process of competition (Hunt and Morgan 1996; Varadarajan 2023; Agnihotri and Gabler 2024). Different from the resource-based view, which emphasizes the importance of VRIN (valuable, rare, inimitable and non-substitutable) resources for gaining competitive advantage, resource-advantage theory highlights the comparative advantage of resources, suggesting that in market environments, there is not just one competitive strategy to achieve superior performance (Varadarajan 2023; Hansen et al. 2023). For instance, firms can pursue different competitive strategies (cost leadership, differentiation, focused cost leadership, or focused differentiation) and market entry strategies (pioneering, early followers, or late entrants) (Varadarajan 2023). Moreover, resource-advantage theory emphasizes that competition is a dynamic process, and firms will evaluate their resources according to changes in the competitive environment to make different strategic choices to obtain relative competitive advantages (Hunt and Morgan 1996).

Based on this theory, we argue that, in the context of horizontal coopetition, the different types and degrees of partner similarity will lead firms to prefer different search strategies for varying purposes. Increased competing similarity can stimulate firms' competitiveness (Runge et al. 2022), driving them to proactive search to seek new technologies different from their partners for a competitive edge. Conversely, greater

cooperating similarity makes it easier for firms to learn from their partners (Park et al. 2014; Vaez-Alaei et al. 2022), encouraging them to choose the low-risk option of responsive search for diverse resources. Therefore, resource-advantage theory is our theoretical support for using partner similarity as a moderating factor.

Overall, as shown in Figure 1, we develop a conceptual model in which proactive and responsive search act as intermediaries in the relationship between horizontal coopetition and SPI. Both competing similarity and cooperating similarity moderate the relationships between horizontal coopetition and each of proactive and responsive search.

2.2 | Research Hypotheses

2.2.1 | Horizontal Coopetition and SPI

We expect that horizontal coopetition affects SPI. Firstly, firms can acquire the resources, critical technologies, and knowledge required for SPI from horizontal coopetition, while its competitive attribute prompts more efficient integration and allocation of resources. SPI is a resource-intensive practice, so it largely depends on extensive resource support (Meiros et al. 2022). Research indicates that initial capital (Hussain et al. 2020), advanced technologies (Rusch et al. 2023), intellectual capital (C. H. Wang and Juo 2021), among others, play crucial roles in SPI. Within horizontal coopetition, firms not only gain diverse innovation resources but also market alignment and similar objectives facilitate easier technology sharing and integration among partners. Secondly, horizontal coopetition provides a mechanism for the collaborating parties to share new resource costs, spread risks, and reduce uncertainties (Corbo et al. 2023). Specifically, SPI is frequently regarded as a high-risk R&D investment, and companies face multiple risks such as poor market response, immature technology, and unclear investment returns (Moshood et al. 2022). In the process of horizontal coopetition, partners integrate their core resources and capabilities to alleviate the financial risk of SPI and reduce uncertainties in the innovation process. Third,

horizontal coopetition can make companies more sensitive to market demand and customer feedback (Chen et al. 2021). In this way, companies can more accurately judge market trends and consumer preferences, enabling them to proactively adjust the design and functionality of sustainable products to meet changing market demands. By combining the preceding considerations, the first hypothesis is as follows:

Hypothesis 1. *Horizontal coopetition is positively correlated with SPI.*

2.2.2 | Horizontal Coopetition and Boundary-Spanning Search

We expect that horizontal coopetition affects boundary-spanning search. On the one hand, according to the coopetition theory, an important mechanism for enterprises to gain benefits from horizontal coopetition is the opportunity to learn from competitors (Xie et al. 2023). By observing competitors' innovative practices and market reactions, and linking their actions and consequences, firms can better understand market dynamics and competitive patterns (Katila and Chen 2008). Therefore, we argue that horizontal coopetition will drive companies to search for new knowledge and resources in response to competitors, which not only reduces the risk and cost of knowledge search but also enhances its effectiveness. On the other hand, horizontal coopetition implies that firms and partners overlap in market positioning, which enhances firms' competitive awareness and motivation (Runge et al. 2022), prompting firms to actively acquire resources that competitors do not possess in order to lead competitors and achieve sustainable competitive advantage. Furthermore, we argue that compared to responsive search, horizontal coopetition has a stronger impact on proactive search. This is because the uniqueness of horizontal coopetition lies in companies experiencing competitive pressure even in cooperation (Raza-Ullah and Kostis 2020). Competition is a race, and gaining innovative resources that competitors lack is crucial; otherwise, they risk losing their competitive edge (Katila and Chen 2008). Accordingly, we suggest the following hypotheses:

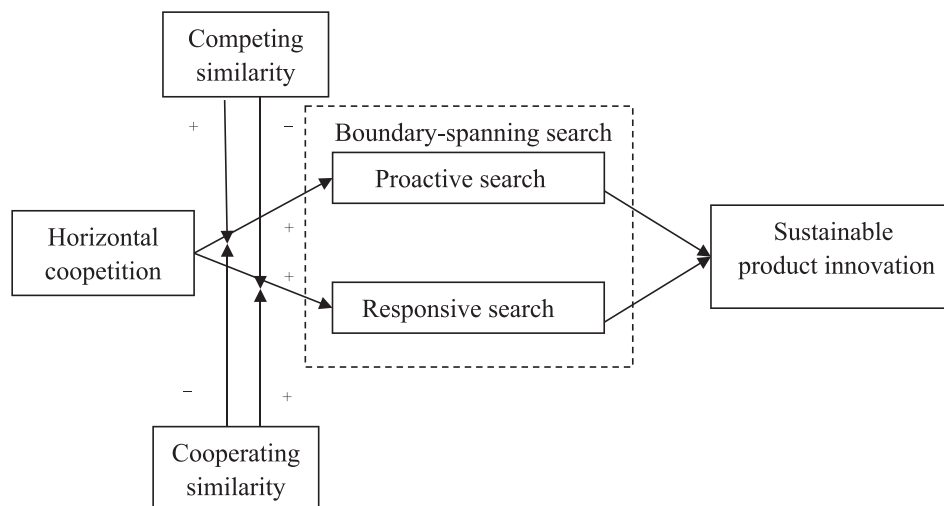


FIGURE 1 | Research model.

Hypothesis 2a. *Horizontal coopetition is positively correlated with responsive search.*

Hypothesis 2b. *Horizontal coopetition is positively correlated with proactive search and has a stronger impact on it compared to responsive search.*

2.2.3 | The Mediating Effect of Boundary-Spanning Search

Horizontal coopetition triggers competitive responses from firms, enabling them to access environmental resources across organizational boundaries (Xin et al. 2023; Li et al. 2024). Meanwhile, complementary resources from outside organizational boundaries open up new possibilities to achieve sustainability (J. Wang et al. 2020). We therefore assume that boundary-spanning search plays a mediating role in the influence of horizontal coopetition on SPI. On the one hand, coopetition theory holds that competition coexists with cooperative activities—even if the degree of competition is very small (Crick et al. 2024). This will induce competitive effects (Park et al. 2014; Rai 2016; Rai and Surana 2022) and drive firms to seek opportunities to surpass competitors and pursue first-mover advantages (Mariani and Belitski 2023). Then they need to engage in proactive search, allowing them to access abundant environmental resources such as eco-friendly materials and green packaging, which provide robust support for firms in developing and designing sustainable products (J. Wang et al. 2020, 2023; Yang et al. 2021). On the other hand, due to similar knowledge bases and expertise, coopetition also stimulates a mutual learning motivation among firms (Park et al. 2014). Therefore, firms may choose to engage in responsive search to acquire market information and innovation resources (Mariani and Belitski 2023). This is because, by observing competitors' practices, innovative products and technologies, responsive search seems more effective in acquiring knowledge (Yang et al. 2021). Efficient knowledge acquisition facilitates the flow, communication, and exchange of knowledge required for SPI (Bossink 2018), thereby enhancing the efficiency of firms' SPI (Liao 2018). As such, we hypothesize the following:

Hypothesis 3a. *Proactive search mediates the relationship between horizontal coopetition and SPI.*

Hypothesis 3b. *Responsive search mediates the relationship between horizontal coopetition and SPI.*

2.2.4 | The Moderating Effects of Competing Similarity

Competing similarity refers to the degree of similarity between firms and partners in terms of geographical market coverage, product market coverage, and technological advantages (Kim and Parkhe 2009). We suggest that the higher the competing similarity, the greater the influence of horizontal coopetition on proactive search. When firms and partners exhibit significant resemblance in market positioning and business models, the pressure from horizontal coopetition accentuates the necessity for differentiated strategies (Runge et al. 2022). This may prompt increased involvement in boundary-spanning search

aimed at outpacing competitors and venturing into new market domains. Conversely, we suggest that competing similarity diminishes the beneficial effects of horizontal coopetition on responsive search. As mentioned earlier, under the drive of horizontal coopetition, firms can acquire market information and innovation resources through responsive search. However, the higher the competing similarity between firms and their partners, the greater the similarity of market pressures and R&D challenges faced by both parties (Kim and Parkhe 2009). So they may be inclined to protect their core technologies and resources from disclosure in order to prevent competitors from gaining a competitive advantage (J. Xiao and Bao 2022). Therefore, knowledge sharing and coordination in the cooperation process will become more difficult; the feasibility of proactive search for enterprises will likely diminish. Accordingly, we propose the following hypotheses:

Hypothesis 4a. *Competing similarity strengthens the positive relationship between horizontal coopetition and proactive search.*

Hypothesis 4b. *Competing similarity weakens the positive relationship between horizontal coopetition and responsive search.*

2.2.5 | The Moderating Effects of Cooperating Similarity

Cooperating similarity refers to the degree of similarity between firms and partners in terms of values, behavioral norms, social or corporate culture, and management practices (Kim and Parkhe 2009). We argue that cooperating similarity strengthens the impact of horizontal coopetition on responsive search. A high level of cooperating similarity implies mutual recognition and similar practices between firms and partners. This facilitates effective communication and coordination, thereby reducing collaboration costs and promoting the transfer and sharing of tacit knowledge such as critical technologies (Oh and Yoo 2022). At the same time, based on similar behavioral norms and values, firms can more effectively coordinate internal and external resources, rapidly understand, absorb, and assimilate new knowledge and technologies. Conversely, the higher the cooperating similarity, the more firms and partners tend to align in terms of values, norms, work processes, employee expectations, and management practices (Kim and Parkhe 2009). This reduces the cognitive gap between the parties, potentially leading to cognitive rigidity and group inertia (Oh and Yoo 2022). In such circumstances, firms may be constrained by established cognitive frameworks when facing the competitive pressure generated by horizontal coopetition. This may hinder the emergence of novel ideas and perspectives in collaboration and restrict firms' exploration of heterogeneous and diversified resources, thereby suppressing boundary-spanning search aimed at surpassing competitors. Accordingly, we suggest the following hypotheses:

Hypothesis 5a. *Cooperating similarity strengthens the positive relationship between horizontal coopetition and responsive search.*

Hypothesis 5b. *Cooperating similarity weakens the positive relationship between horizontal coopetition and proactive search.*

3 | Methodology

3.1 | Samples and Data Collection

This study employed a questionnaire survey to collect data, and the content validity of the questionnaire was based on existing literature. We chose the manufacturing industry as the research sample because it plays a predominant role in driving China's economic advancement and has the highest carbon dioxide (CO₂) emissions among industrial sectors (Zhao et al. 2017). Hence, under China's climate crisis response policies, there exists an urgent necessity for SPI within the manufacturing industry to facilitate the transition from extensive growth to sustainable development.

We developed an English language questionnaire based on existing literature and enlisted two independent translators to translate the questionnaire into Chinese language and perform back translation. Prior to the official survey, a pilot survey was conducted with 30 companies in the Chinese Shaanxi Province, and the questionnaire content was adjusted accordingly. To ensure representative sampling, we systematically selected firms across various regions of China, including the northwestern, southern, eastern, and central areas. With the help of established partnerships with local governments and industry associations, we obtained local business directories for each region and randomly chose 250 firms from the local business directory within each of the four regions, totaling 1000 firms. Subsequently, professional interviewers were hired to contact the senior managers of these companies, either by phone or email, explaining the academic objectives of the survey and the confidentiality policy. A total of 530 firms consented to participate in this study. Subsequently, after scheduling appointments, our interviewers visited the companies to conduct the surveys. During the completion of the questionnaires, we thoroughly described the process and answered any questions posed by the respondents. Each company's survey took an average of about 1h to complete.

To avoid potential common method bias, we employed a multiple-informant approach to collect our data (Podsakoff et al. 2012). First, we selected two senior managers from each company to serve as our key informants. These senior managers possess extensive knowledge about their firms' interfirm cooperative activities and innovation performance. Two senior managers from each company were instructed to focus on their most important partner within the past 3 years and provide the name of the cooperating company and the duration of the cooperation. Then, one manager provided responses to basic company information, independent variables (e.g., horizontal coopetition, proactive search, and responsive search), and control variables (e.g., firm age, size), while the scales of the moderating variables (competing similarity, cooperating similarity) and the dependent variable (SPI) were assessed by another manager.

In total, 321 questionnaires were returned, and after rigorous screening, 85 invalid questionnaires were excluded, resulting in a total of 236 responses received. The exclusion of invalid questionnaires was due to data missing content and obvious errors in common sense content. The final sample included

companies from 18 cities in China, primarily concentrated in provinces and cities such as Shaanxi, Jiangsu, Henan, and Guangzhou. The industries mainly comprised manufacturing sectors including construction materials, automotive and parts, environmental protection and landscaping, as well as clothing and textiles. It consisted of 67 state-owned enterprises, 107 private enterprises, and 62 other types of enterprises.

3.2 | Measures

A mature 7-point Likert scale was used for all the scale items (Table 1), with 1–7 indicating a range from *Strongly disagree* to *Strongly agree*.

3.2.1 | Horizontal Coopetition (HC)

Based on Bouncken and Kraus (2013), we used three items (Cronbach's $\alpha=0.788$) to measure horizontal coopetition. An example item is "Our company collaborates with direct competitors to achieve common goals."

3.2.2 | Boundary-Spanning Search

Drawing on the research of Katila and Chen (2008), as well as Atuahene-Gima and Murray (2007), we used four items (Cronbach's $\alpha=0.953$) to measure *proactive search* (PS). An example item is "Our company prefers to collect information with no identifiable strategic market needs to ensure experimentation in the project, aiming to surpass competitors" in boundary-spanning search activities. Additionally, we used four items (Cronbach's $\alpha=0.824$) to measure *responsive search* (RS). An example item is "Our aim is to search for information to refine common methods and ideas in solving problems in the project in order to follow competitors" in boundary-spanning search activities.

3.2.3 | Partner Similarity

Based on Kim and Parkhe (2009) and Zhang et al. (2010), we utilized four items (Cronbach's $\alpha=0.872$) to measure *competing similarity* (CMS). An example item is "Our company has the same production line as our partner." Drawing on the research of Geringer (1991) and Kale et al. (2000), we used three items (Cronbach's $\alpha=0.824$) to measure *cooperating similarity* (COS). An example item is "Our company shares the same organizational culture with our partner."

3.2.4 | Sustainable Product Innovation (SPI)

Drawing on the research of Klewitz and Hansen (2014) and Peng and Lin (2008), the five items (Cronbach's $\alpha=0.884$) were used to measure SPI. An example item is "Our company develops products according to the product lifecycle method."

To control for potential influences from other variables that may affect the study results, we included the following control

TABLE 1 | Measures and validation.

Constructs	Loading
Horizontal coopetition	
Our company maintains a close competitive relationship with the partner.	0.789
Our company collaborates with direct competitors to achieve common goals.	0.676
Our company recognizes that maintaining effective competition with our partner is crucial for our success.	0.772
Proactive search	
To what extent do you rate your company's boundary-spanning search activities?	
In our search, our company focuses on acquiring knowledge of project strategies that involve experimentation and high market risks in order to surpass competitors.	0.917
Our company prefers to collect information with no identifiable strategic market needs to ensure experimentation in the project, aiming to surpass competitors.	0.936
Our company collects novel information and ideas that went beyond our current market and technological experiences.	0.897
Our aim was to acquire new resources that forced us to learn new things in the product development project, surpassing competitors.	0.906
Responsive search	
Our aim is to acquire resources to refine common methods and ideas in solving problems in the project in order to follow competitors	0.774
Our aim is to search for ideas and information that we can implement well to ensure productivity to keep pace with competitors rather than those ideas that could lead to implementation mistakes in the project and in the marketplace.	0.817
Our company searches for the usual and generally proven methods and solutions to product development problems in order to keep up with competitors.	0.696
Our company obtains resources through surveys of current customers and competitors, which helped us understand and update the firm's current project and market experiences to keep up with competitors.	0.645
Competing similarity	
Our company have the same suppliers as our partner.	0.751

(Continues)

TABLE 1 | (Continued)

Constructs	Loading
Our company are in the same product market as the partner.	0.850
Our company have a product line very similar to the partner's	0.813
Our company need the same type of knowledge related to new product or process development as the partner's.	0.765
Cooperating similarity	
Our company's organizational culture is compatible with that of our partner.	0.832
Our company's management and operating styles are compatible with those of our partner.	0.845
Our company and the partner maintain relatively high mutual trust in various activities.	0.773
Sustainable product innovation	
Our company develops products according to the product lifecycle method.	0.701
Our company adopts eco-friendly product design.	0.842
Our company applies green product promotion and advertising methods.	0.851
Our company strives to use recyclable packaging materials in our products.	0.778
Our company implements waste recycling processes for our products.	0.726
Environmental turbulence	
Customer preferences for products change rapidly over time.	0.644
Customers always tend to pursue new products.	0.623
In our industry, technological changes occur very quickly.	0.774
This industry can develop many new products through technological breakthroughs.	0.726
The frequency of technological changes is high.	0.816
Environmental competitiveness	
The local market competition is intense.	0.868
Our company faces many strong competitors.	0.751
The local market competition is very high.	0.892
Price competition is a characteristic of the local market.	0.638

variables in our analysis: firm age, size, cooperation duration, environmental turbulence, and environmental competitiveness. Firm age and cooperation duration were measured as the

TABLE 2 | Descriptive statistics, Pearson correlations, reliability, and validity.

Constructs	STD. loading	CR	AVE	ET	EC	HC	PS	RS	CMS	COS	SPI
ET	0.623– 0.816	0.842	0.519	0.720							
EC	0.638– 0.892	0.870	0.630	0.413***	0.794						
HC	0.676– 0.789	0.791	0.559	0.058	−0.058	0.748					
PS	0.897– 0.936	0.953	0.836	0.176*	0.047	0.334***	0.914				
RS	0.645– 0.817	0.824	0.542	0.263**	−0.125 [†]	0.371***	0.177*	0.736			
CMS	0.751– 0.850	0.873	0.633	0.222**	0.196**	0.421***	0.252***	0.094	0.796		
COS	0.773– 0.845	0.858	0.668	0.129 [†]	−0.018	0.202*	0.063	0.427***	0.137 [†]	0.817	
SPI	0.701– 0.851	0.887	0.611	0.184*	0.194*	0.447***	0.299***	0.452***	0.145 [†]	0.319***	0.782
Mean				4.448	5.083	4.400	4.515	4.354	4.389	4.905	5.125
SD				1.079	1.151	1.075	1.462	0.961	1.127	0.827	0.935

Note: $N = 236$. The square roots of AVE appear on the diagonal. Nondiagonal elements (nonbold) are the correlations among constructs.

Abbreviation: SD = standard deviation.

***indicates the significance level $p < 0.001$.

**indicates the significance level $p < 0.01$.

*indicates the significance level $p < 0.05$.

[†]indicates the significance level $p < 0.1$.

natural logarithm of the firm's age and partnership period, respectively. Firm size was transformed using the logarithm of the number of employees with a base of 10. *Environmental turbulence* (ET) was measured using the scale developed by DeSarbo et al. (2005), which consisted of five items (Cronbach's $\alpha = 0.842$). *Environmental competitiveness* (EC) was measured using the scale developed by Jansen et al. (2006), which consisted of four items (Cronbach's $\alpha = 0.866$).

3.3 | Reliability and Validity

As shown in Table 2, the reliability and validity analysis of the scales yielded favorable results. The Cronbach's α coefficients and composite reliabilities (CR) of all variables were greater than 0.7, indicating a high reliability of the questionnaire. Furthermore, the standardized factor loadings of each variable ranged from 0.6 to 0.9, and the average variance extracted (AVE) values were greater than 0.5, indicating that the measure had good convergent and discriminant validity of the variables used in this study. At the same time, the square root of the AVE (indicated by bold numbers on the diagonal) was greater than the correlation coefficients between the respective variable and other variables, indicating that the variables have good discriminant validity.

3.4 | Common Method Variance

We adopted two methods for checking common method variance (CMV) in this study. First, Harman's single-factor method was used to analyze the influence of CMV. The results revealed that the first factor's variance contribution, in the unrotated factor solution, was 21.137%, which was less than the recommended threshold of 40%. This indicated that the CMV in this study was within an acceptable range. Second, we adopted confirmatory factor analysis to evaluate Harman's single-factor model. Specifically, the model fit based on the single-factor model ($\chi^2 = 3803.225$, $df = 464$, $\chi^2/df = 8.197$, $p < 0.001$, CFI = 0.247, TLI = 0.195, NFI = 0.229, RMSEA = 0.175, and PGFI = 0.381) was inferior to that of the eight-factor model ($\chi^2 = 822.118$, $df = 436$, $\chi^2/df = 1.886$, $p < 0.001$, CFI = 0.913, TLI = 0.901, NFI = 0.833, RMSEA = 0.061, and PGFI = 0.680). Therefore, the CMV was not an issue in this study.

3.5 | Descriptive Statistics and Correlations

Table 2 presented the results of mean, standard deviation, and variable correlation results for the variables included in this study. The correlation coefficients indicated that there were significant relationships among the variables, and the

TABLE 3 | Multiple regression results.

Variables	PS			RS			SPI		
	M1	M2	M3	M4	M5	M6	M7	M8	M9
Controls									
Firm size	0.064	0.067	0.093	0.068	0.070	0.092	0.095	0.097	0.069
Firm age	−0.097	−0.118	−0.084	−0.011	−0.021	−0.053	−0.055	−0.071	−0.055
Cooperation duration	−0.018	0.092	0.036	−0.040	0.014	0.044	0.005	0.089	0.077
ET	0.226*	0.187*	0.115 [†]	0.259***	0.240***	0.245***	0.099	0.070	−0.022
EC	−0.015	0.026	−0.024	−0.182**	−0.162**	−0.130*	0.097 [†]	0.128*	0.178**
Independent variable									
Horizontal cooperation (HC)		0.408***	0.408***		0.199***	0.121*		0.313***	0.216***
Moderating variables									
Competing similarity (CMS)			0.166 [†]			−0.056			
Cooperating similarity (COS)			−0.087			0.407***			
Interactions									
HC*CMS			0.166**			−0.111**			
HC*COS			−0.158*			0.156***			
Mediating variable									
PS									0.083*
RS									0.318***
R ²	0.030	0.116	0.167	0.086	0.134	0.279	0.047	0.171	0.283
ΔR ²	0.030	0.086	0.138	0.086	0.047	0.193	0.047	0.124	0.236
F	1.415 [†]	5.000***	4.525***	4.352***	5.889***	8.712***	2.263*	7.869***	11.211***

Note: N = 236.

***indicates the significance level $p < 0.001$.

**indicates the significance level $p < 0.01$.

*indicates the significance level $p < 0.05$.

[†]indicates the significance level $p < 0.1$.

magnitude and direction of the correlations align with the research expectations. Specifically, horizontal cooperation was significantly positively correlated with SPI ($r = 0.447$, $p < 0.001$), proactive search ($r = 0.334$, $p < 0.001$), and responsive search ($r = 0.371$, $p < 0.001$). These findings provided preliminary support for the validity of the model and hypotheses, indicating the need for further examination of the relationships among the variables.

4 | Results

4.1 | Direct Effects of Horizontal Cooperation

We tested the hypothesis using hierarchical regression analysis, and centralization was conducted to the independent and moderating variables to avoid multicollinearity concerns. According to Model 8, we found that horizontal cooperation had

a significantly positive impact on SPI ($\beta = 0.313$, $p < 0.001$), as shown in Table 3. Therefore, Hypothesis 1 was supported. In Models 2 and 5, the results indicated the positive influence of horizontal cooperation on proactive search ($\beta = 0.408$, $p < 0.001$) and responsive search ($\beta = 0.199$, $p < 0.001$). Then, a Z-test was used to examine whether horizontal cooperation has statistically significant differences in the impact of proactive search and responsive search. The results showed that horizontal cooperation had a stronger impact on proactive search than on responsive search ($Z\text{-value} = 2.037 > 1.96$, $p < 0.05$). Therefore, Hypotheses 2a and 2b were supported.

4.2 | Mediations Effects of Boundary-Spanning Search

To test for the mediating effect, we included the mediator variables proactive search and responsive search in Model 8,

TABLE 4 | Multiple mediation analysis results.

Path	Point estimate value	Bootstrapping		
		Bias-corrected		
		Confidence intervals		
		SE	Lower	Upper
Horizontal coopetition → PS → SPI	0.065	0.028	0.024	0.034
Horizontal coopetition → RS → SPI	0.040	0.018	0.010	0.083

Note: $N = 236$.

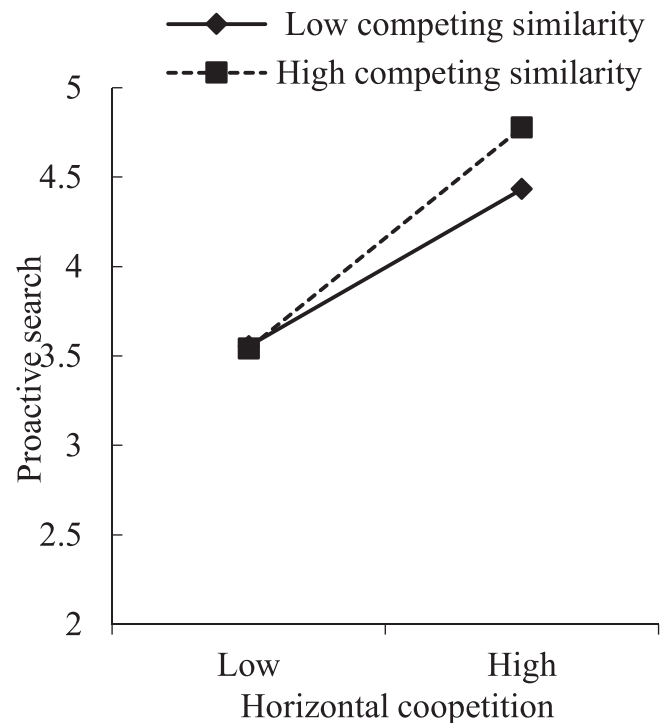
Abbreviation: SE = standard errors.

resulting in Model 9. We found that the relationship between proactive search and SPI was positive and significant ($\beta = 0.083$, $p < 0.05$) and the relationship between responsive search and SPI was also positive and significant ($\beta = 0.318$, $p < 0.001$). At the same time, the direct positive impact of horizontal coopetition on SPI ($\beta = 0.216$, $p < 0.001$) remained significant but weakened. Therefore, both proactive and responsive search partially mediated the relationship between horizontal coopetition and SPI, supporting Hypotheses 3a and 3b. To ensure the stability of the mediation test, we used a bootstrap approach to further verify the mediating effects. The results are summarized in Table 4. With a 95% bias-corrected confidence interval (CI) for 1000 repetitions, the indirect effect of horizontal coopetition on SPI via proactive search was significant (effect = 0.065, 95% CI = [0.024, 0.034]); the indirect effect of horizontal coopetition on SPI via responsive search was also significant (effect = 0.040, 95% CI = [0.010, 0.083]). Therefore, Hypotheses 3a and 3b were further supported.

4.3 | Moderating Effects of Partner Similarity

The results of the moderation effects were presented in Models 3 and 6. The coefficient of the interaction term of horizontal coopetition and competing similarity was positive and marginally significant for proactive search (Model 3, $\beta = 0.166$, $p < 0.01$), while negative and marginally significant for responsive search (Model 6, $\beta = -0.111$, $p < 0.01$). Therefore, Hypotheses 4a and 4b were supported. Conversely, the coefficient of the interaction term of horizontal coopetition and cooperating similarity was positive and marginally significant for proactive search (Model 6, $\beta = 0.156$, $p < 0.001$), while negative and marginally significant for responsive search (Model 3, $\beta = -0.158$, $p < 0.05$). Therefore, Hypotheses 5a and 5b were supported.

To gain further insights into these interaction relationships, we plotted the moderating effects of different partner similarities in Figures 2–5. Figure 2 illustrates that the positive linear relationship between horizontal coopetition and proactive search has an upward trend when competing similarity is greater. As shown in Figure 3, the positive linear relationship between horizontal coopetition and responsive search has a downward trend when competing similarity is greater. Figure 4 illustrates that the positive linear relationship between horizontal coopetition and responsive search has an upward trend when cooperating similarity is greater.

**FIGURE 2** | Moderating effect of competing similarity between horizontal coopetition and proactive search.

As shown in Figure 5, the positive linear relationship between horizontal coopetition and proactive search has a downward trend when cooperating similarity is greater.

5 | Discussion

5.1 | Key Findings

The background of the study is that coopetition is an important factor in firms' strategies to achieve sustainability. The objective of our study is to investigate whether and how horizontal coopetition affects firms' SPI. We empirically examined the mediating role of boundary-spanning search in the relationship between horizontal coopetition and SPI, as well as the moderating effects of partner similarity. The results of the empirical tests are shown in Table 5. Below, we provide a detailed discussion of the research findings in this paper.

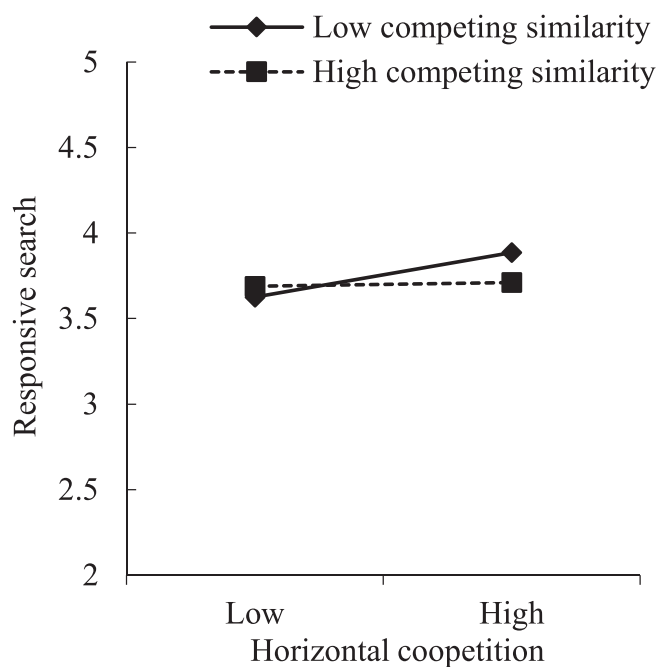


FIGURE 3 | Moderating effect of competing similarity between horizontal cooperation and responsive search.

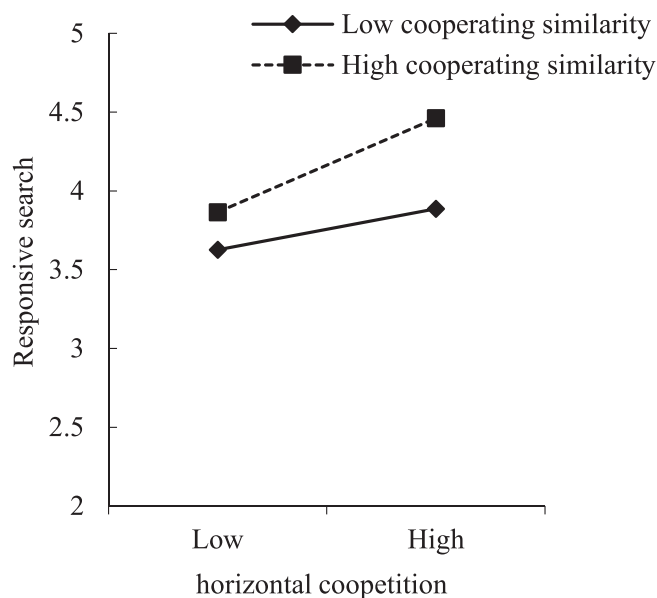


FIGURE 4 | Moderating effect of cooperating similarity between horizontal cooperation and responsive search.

Regarding Hypothesis 1, our research findings indicate that horizontal cooperation has a significant positive impact on SPI. This finding is similar to the findings of Mwesumio et al. (2023), Vu and Fath (2024), and Xin et al. (2023); that is, cooperation has a positive impact on corporate sustainability. As Corbo et al. (2023) argue, although there are potential drawbacks in cooperation, an abundance of empirical and anecdotal evidence supports the rationale that cooperation benefits innovation-related activities. Our research focuses on the context of corporate sustainability and confirms that

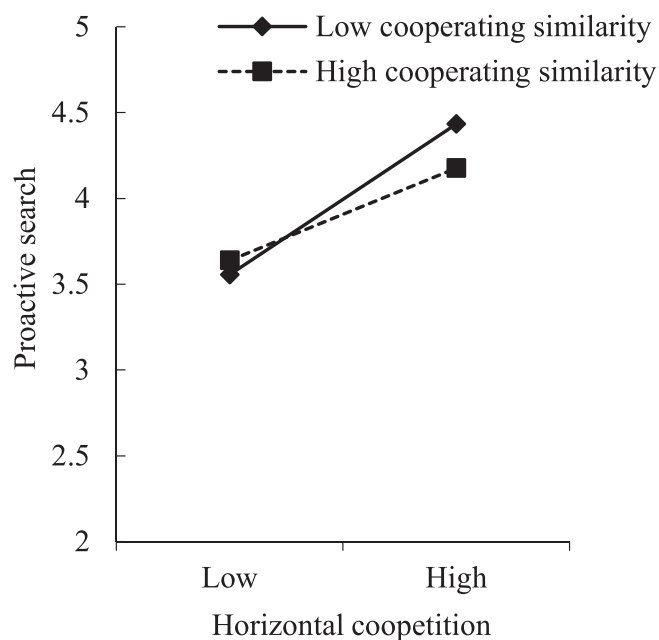


FIGURE 5 | Moderating effect of cooperating similarity between horizontal cooperation and proactive search.

horizontal cooperation has a promoting effect on SPI. That is, collaboration with companies of similar market positioning or technological expertise not only enhances the environmental performance and economic efficiency of new products but also generates social benefits. This is a crucial result, since it offers empirical support to scholars' calls for investigating the role of competition in sustainable innovation (Munten et al. 2021; Corbo et al. 2023; Xin et al. 2023).

Concerning Hypothesis 2, we found that horizontal cooperation positively impacts both proactive and responsive search. As Shi et al. (2024) pointed out, there is a significant lack of research on the antecedents of boundary-spanning search. Our study confirms horizontal cooperation not only stimulates enterprises to engage in boundary-spanning search with the purpose of surpassing competition, but also stimulates enterprises to conduct such searches in response to competitors. This finding also supports the research by Shi et al. (2024) and Ortiz et al. (2018) that firm partnerships are a prerequisite for external knowledge search. More interestingly, we found that horizontal cooperation has a stronger impact on proactive search than on responsive search. Our explanation for this is that in the current competitive environment where customer preferences change rapidly and technological advancements are significant, if companies want to benefit from horizontal cooperation, they must quickly launch products (Nemeh and Yami 2019). This further promotes the need for companies to proactively search for resources that are significantly different from their competitors, thus gaining an advantageous position in the market.

In Hypothesis 3, both proactive and responsive search play a mediating role in the relationship between horizontal cooperation and SPI. Although existing research on the mediating effects of horizontal cooperation on SPI is scarce, studies related to how competition influences firms' innovation, such as that of Zhang

TABLE 5 | Summary of hypotheses and empirical results.

No.	Hypothesis	Supported (yes/no)
Hypothesis 1	Horizontal coopetition is positively correlated with SPI.	Yes
Hypothesis 2a	Horizontal coopetition is positively correlated with responsive search.	Yes
Hypothesis 2b	Horizontal coopetition is positively correlated with proactive search and has a stronger impact on it compared to responsive search.	Yes
Hypothesis 3a	Proactive search mediates the relationship between horizontal coopetition and SPI.	Yes
Hypothesis 3b	Responsive search mediates the relationship between horizontal coopetition and SPI.	Yes
Hypothesis 4a	Competing similarity strengthens the positive relationship between horizontal coopetition and proactive search.	Yes
Hypothesis 4b	Competing similarity weakens the positive relationship between horizontal coopetition and responsive search.	Yes
Hypothesis 5a	Cooperating similarity strengthens the positive relationship between horizontal coopetition and responsive search.	Yes
Hypothesis 5b	Cooperating similarity weakens the positive relationship between horizontal coopetition and proactive search.	Yes

et al. (2010), have identified that knowledge acquisition and knowledge creation can help firms achieve greater innovation performance through coopetition; Chen et al.'s (2021) research has revealed the mediating role of external knowledge integration in the relationship between coopetition and radical innovation. Consistent with these findings, our study also highlights the importance of knowledge acquisition in the process of coopetition influencing innovation by investigating the mediating role of boundary-spanning search.

Regarding Hypothesis 4, cooperating similarity enhances the positive impact of horizontal coopetition on responsive search, while it weakens the positive impact of horizontal coopetition on proactive search. This is consistent with the perspective of Bacon et al. (2020) and Gao et al. (2017), who argue that consistency among participants in coopetition regarding shared beliefs, values, and management practices facilitates tacit knowledge transfer. This will further promote firms' learning from competitors, thereby strengthening the impact of horizontal coopetition on responsive search. At the same time, this may lock participants into a set of decision-coordination norms, increasing transaction costs and preventing any party from independently launching new market products (Mariani and Belitski 2023), thereby weakening the impact of horizontal coopetition on proactive search.

In Hypothesis 5, competing similarity strengthens the positive impact of horizontal coopetition on proactive search while weakening its positive impact on responsive search. This also confirms the view of Luo (2007) and Farazi et al. (2024) that within coopetitive relationships, the higher the competing similarity, the more challenging it is for a company to obtain a larger market share and gain a competitive advantage over its competitors. This further strengthens the company's tendency to gain a competitive advantage through boundary-spanning search aimed at surpassing its competitors. Simultaneously, as Ritala and Hurmelinna-Laukkanen (2009) and Rouyre et al. (2024) pointed out, competing similarity also intensifies opportunistic

behavior, so that prompt participants to establish mechanisms to prevent imitation and knowledge theft. This, in turn, inhibits firms' pursuit of boundary-spanning search aimed at following competitors.

5.2 | Theoretical Implications

The current study has several theoretical implications. First, by revealing the relationship between horizontal coopetition and SPI, this research expands the current body of literature on the relationship between coopetition and firms' sustainability. Recently, the significance of coopetition in firms' sustainability has gradually attracted scholars' attention (Planko et al. 2019; Munten et al. 2021; Corbo et al. 2023; Mwesiumo et al. 2023). However, SPI, as a crucial pathway for firms to achieve sustainability, has been seldom empirically supported for its association with horizontal coopetition. Therefore, this study fills the gap in empirical research on the value of horizontal coopetition in enhancing SPI and also serves as a beneficial supplement to the study of preconditions for SPI.

Second, by innovatively introducing boundary-spanning search into the research framework of the relationship between horizontal coopetition and SPI, this study expands and enriches the research on the underlying mechanisms between them. Mwesiumo et al. (2023) pointed out the lack of exploration into the underlying mechanisms through which coopetition affects firms' sustainability. The scholars argued that the lack of exploration into the underlying mechanisms hinders the advancement of coopetition theory and also impedes our understanding of the value creation process of coopetition. Therefore, by introducing boundary-spanning search, this study not only bridges the gap between horizontal coopetition and SPI but also advances the development of coopetition theory.

Third, this study encourages the integration of coopetition theory and resource advantage theory. Coopetition theory posits

that coopetition has simultaneity, meaning competition and cooperation coexist simultaneously (Corbo et al. 2023). In other words, firms make different strategic choices to gain competitive advantages in coopetition. However, coopetition theory has overlooked the crucial role of partner similarity in strategic decision-making under coopetition. Resource-advantage theory, on the other hand, explains this phenomenon by suggesting that firms make different strategic choices based on their relative resource advantages in dynamic competitive environments (Varadarajan 2023). This study integrates these two perspectives, validating that in horizontal coopetition, enterprises will judge their relative resource advantages based on similarity with their partners, and will tend to make different boundary-spanning search strategies based on this. These findings expand the boundaries of current research on how horizontal coopetition influences firm behavior, provide a beneficial supplement to coopetition theory, and respond to the question raised by Ehls et al. (2020): Under what conditions is one external search activity more valuable than another during the innovation phase.

5.3 | Practical Implications

The present study has several practical implications. First, given the significant role of horizontal coopetition in SPI, managers should maintain an open mindset and actively identify and establish cooperative relationships with trustworthy competitors. Managers should realize that, despite firms being in competition with direct rivals in the market, the value created through collaboration with a specific competitor in today's vast and complex market environment can be utilized to address challenges posed by other competitors. Particularly in the context of SPI, companies face issues such as scarcity of resources and technological deficits, while also considering risk factors. Competitors often encounter similar dilemmas; therefore, collaboration not only provides access to resources but also enables joint risk mitigation. Additionally, business managers should realize that horizontal coopetition directly drives boundary-spanning search activities within companies. During the process of cooperating with competitors, managers should carefully observe and learn from competitors' management practices and operational strategies, drawing experiences and lessons to select appropriate boundary-spanning search strategies based on their own circumstances. Whether conducting boundary-spanning search strategies aimed at surpassing competitors or following competitors' lead, managers should closely monitor competitors' dynamic changes, establish effective information retrieval mechanisms, and better formulate boundary-spanning search strategies.

Second, our study offers new potential pathways for companies to achieve SPI through horizontal coopetition. The research finds that horizontal coopetition can positively influence SPI through both proactive and responsive search. On the one hand, driven by horizontal coopetition, managers should value the role of responsive search aimed at following competitors in SPI. By observing and learning from competitors' innovative practices and experiences, managers can better understand and grasp market dynamics and industry trends. Thus, managers can reduce the trial-and-error costs and risks in SPI by engaging in responsive search aimed at following competitors, thereby increasing the likelihood of successful SPI. On the other hand, driven by horizontal

coopetition, proactive search aimed at surpassing competitors can similarly foster SPI. Managers should encourage teams to think beyond traditional frameworks and actively explore innovative opportunities surpassing competitors. By conducting search across organizational boundaries to explore new technologies and concepts from different industries or domains, new perspectives can be provided for SPI, enhancing product differentiation and further strengthening the company's competitiveness.

Finally, our study suggests that managers should clearly define selection criteria when establishing cooperative relationships with competitors and choose different types of partners based on different strategic objectives. On the one hand, if a company intends to engage in boundary-spanning search aimed at following competitors in the future, managers should investigate and evaluate the similarities in culture, values, and management practices of potential partners. They should actively establish good cooperation with companies with high similarity. Cooperating similarity not only enhances the stability of cooperation but also makes it easier for both parties to understand each other's expectations and behavioral norms, facilitating knowledge sharing and learning. This accelerates the understanding and absorption of competitor practices, thereby achieving a more effective boundary-spanning search aimed at following competitors. On the other hand, if a company intends to engage in boundary-spanning search aimed at surpassing competitors in the future, managers should choose companies with high similarity in competitive aspects such as geographical market coverage, product market coverage, and technological advantage areas. The higher the competing similarity, the more similar the company's market positioning and resource allocation are with the partner. In this coopetition relationship, managers should have a deep understanding of competitors' product advantages and market competition situations and formulate strategies more targeted to surpass competitors.

5.4 | Limitations and Future Research

Although this study explores the mechanism of horizontal coopetition from both theoretical and empirical perspectives, some research limitations still need to be further investigated and improved in future research. First, our sample is limited to manufacturing companies and does not include other industries engaged in SPI. Moreover, since our research results are based on samples from Chinese companies, the generalizability of the research conclusions may be limited. Therefore, in future research, these findings should be validated in other countries and industries.

Second, although we implemented a multiple-informant approach when collecting questionnaires to reduce common method bias, the subjective judgments of the respondents inevitably lead to some degree of common method bias in the data analysis process. Therefore, in future research, scholars should adopt a combination of subjective and objective data to avoid this issue. Additionally, the cross-sectional data used in this study may not fully represent dynamic causal conclusions. Therefore, future research can avoid this issue by tracking data from multiple time points.

Third, the outcomes of sustainable innovation are not only reflected in product development but also in processes. Therefore,

future research can incorporate sustainable process innovation into the research model to explore the differences in the impact of horizontal coopetition on SPI and sustainable process innovation, thereby gaining a deeper understanding of the role of horizontal coopetition in firms' sustainability development.

Fourth, this study confirms the mediating role of boundary-spanning search in the relationship between horizontal coopetition and SPI. Future research can explore the mechanism of how horizontal coopetition affects SPI based on other theoretical perspectives, such as knowledge management theory, thereby furthering understanding of the underlying mechanisms through which horizontal coopetition affects SPI.

6 | Conclusion

This study is based on coopetition theory and constructs a theoretical model of "horizontal coopetition—boundary-spanning search (proactive/responsive)—SPI." Building upon this framework, the study integrates resource-advantage theory to analyze the moderating effect of partner similarity (cooperating/competing) on the relationship between horizontal coopetition and boundary-spanning search. Using survey data from 236 Chinese manufacturing firms, the results indicate that horizontal coopetition has a positive impact on SPI. Furthermore, horizontal coopetition positively influences both proactive and responsive search, with a stronger effect on proactive search compared to responsive search. Additionally, our findings suggest that both proactive and responsive search mediate the relationship between horizontal coopetition and SPI. Finally, we reveal the moderating effect of partner similarity on the relationship between horizontal coopetition and boundary-spanning search. Specifically, competing similarity strengthens the positive effect of horizontal coopetition on proactive search but weakens the positive effect on responsive search; while cooperating similarity strengthens the positive effect of horizontal coopetition on responsive search but weakens the positive effect on proactive search.

Author Contributions

Xia Wu: writing – review and editing, writing – original draft, formal analysis, investigation, methodology, conceptualization. **Jianjun Yang:** formal analysis, investigation, methodology, supervision, conceptualization, final approval. **Sandra Hasanefendic:** writing – review and editing, supervision, conceptualization. **Bart Bossink:** writing – review and editing, supervision, conceptualization.

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Conflicts of Interest

The authors declare no conflicts of interest.

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